Reg. No															

B.Tech/M.Tech(Integrated) DEGREE EXAMINATION, DECEMBER 2023

Third Semester

21CSC201J - DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted during the academic year 2022-2023 onwards)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

		Marks	BL	CO
1.	We can certainly get a conclusion about an algorithm's _____ scenario from asymptotic analysis. (A) best case (B) best case, average case, and worst case (C) worst case (D) average case	1	1	1
2.	If 'p' is declared as integer pointer, then, an array 'a' can be pointed by the following assignment (A) p = &a[0] (B) p++ (C) p == &a[0] (D) p == a[0]	1	1	1
3.	Which of the following is the correct declaration of self-referential structure? (A) struct node*{int data;node *link;} (B) struct node {int data; struct node *link; }; (C) struct node{int data;node *link;} (D) struct node* {int data; struct node *link; }	1	2	1
4.	How do you initialize an array in C? (A) int arr[3] = (1,2,3); (B) int arr(3) = {1,2,3}; (C) int arr[3] = {1,2,3}; (D) int arr(3) = (1,2,3);	1	1	1
5.	Which of the following c code is used to create a new node? (A) ptr=(NODE*)malloc(sizeof(NODE)); (B) ptr=(NODE*)malloc(NODE); (C) ptr=(NODE*)malloc(sizeof(NODE*)); (D) ptr=(NODE)malloc(sizeof(NODE));	1	2	2
6.	Which of the following is a practical example of a doubly linked list? (A) A browser cookie file (B) A quest in a game that lets users retry stages. (C) A game in which the player runs forward. (D) A first-in-first out scheduling system.	1	1	2
7.	Polynomial addition can be implemented using which of the following data structures? (A) Linked List (B) Trees (C) Stacks (D) Priority Queue	1	1	2
8.	In a Circular linked list organization, insertion of a record involves the modification of _____ (A) No pointer (B) 1 pointer (C) 2 pointer (D) 3 pointer	1	1	2

9. The following sequence of operations is performed on stack: PUSH (30),PUSH (40),POP,PUSH (30),PUSH(40),POP,POP,POP,PUSH (40),POP. The sequence of the value popped out is: 1 3 3

(A) 40,30,40,30,40 (B) 30,40,40,30,40
(C) 40,40,30,40,30 (D) 40,40,30,30,40

10. The_____ operation displays the topmost value but will not delete it from the stack 1 1 3

(A) Peek (B) Push
(C) Pop (D) Enqueue

11. A circular queue is implemented using an array of size 10. The array index starts with 0, front is 6, and rear is 9. The insertion of next element takes place at the array index. 1 2 3

(A) 0 (B) 7
(C) 9 (D) 10

12. Which is the most appropriate data structure for reversing a word? 1 2 3

(A) queue (B) stack
(C) tree (D) graph

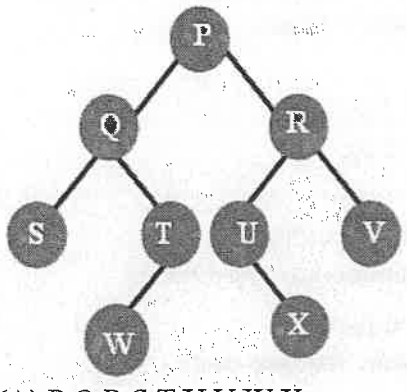
13. In Binary tree traversing first visit the root node, then left sub tree and right sub tree is called_____ 1 1 4

(A) Inorder (B) Preorder
(C) Postorder (D) Levelorder

14. What is the formula used in quadratic probing? 1 1 4

(A) Hash key = key mod table size (B) Hash key=(hash(x)+F(i)) mod table size
(C) Hash key=(hash(x)+F(i^2)) mod table size (D) H(x) = x mod 17

15. Find the postorder traversal of the binary tree shown below 1 2 4

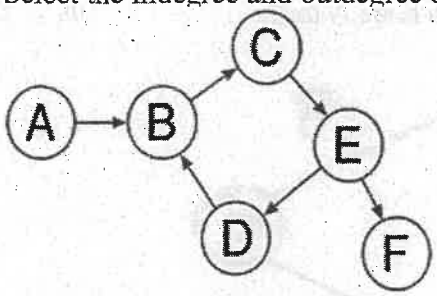


(A) P Q R S T U V W X (B) W R S Q P V T U X
(C) S W T Q X U V R P (D) S T W U X V Q R P

16. Select the correct definition for balancing factor of a tree 1 1 4

(A) Difference between left subtree and right subtree (B) Difference between leaf node and non-leaf node
(C) Difference between root and degree of a tree (D) Difference between height and depth of a tree

17. Select the indegree and outdegree of a node E for the given graph. 1 2 5



(A) Indegree 1 and Outdegree 1 (B) Indegree 2 and Outdegree 2

(C) Indegree: 0, Outdegree: 2 (D) Indegree: 1, Outdegree: 3

18. A person wants to visit some places. He starts from a vertex and then wants to visit every vertex till it finishes from one vertex, backtracks, and then explores another vertex from the same vertex. What algorithm he should use? 1 2 5

(A) Depth First Search (B) Breadth-First Search

(C) Trim's algorithm (D) Kruskal's Algorithm

19. Identify the number of edges that a completed graph with 'n' vertices have? 1 2 5

(A) (n*(n-1))/2 (B) (n*(n+1))/2

(C) n (D) 2n

20. A man wants to go different places in the world. He has listed them down all. But there are some places where he wants to visit before some other places. What application of graph can he use to determine that? 1 2 5

(A) Depth First Search (B) Breadth First Search

(C) Topological Sorting (D) Dijkstra's Shortest path algorithm

PART - B (4 × 10 = 40 Marks) Marks BL CO

Answer **any 4** Questions

21. a) Find the tight bound of running time of a cubic function. $f(n) = 2n^3+3n+6$; $g(n) = n^3$ 10 3 1

b) Prove that $10n^3 + 20n \neq O(n^2)$.

22. Consider a goods train, which initially has an engine as a header. Later compartments are linked with engines one by one in series. The compartments can be attached or detached with the engine or in between. Implement a data structure for the following operations in the goods train 10 3 2

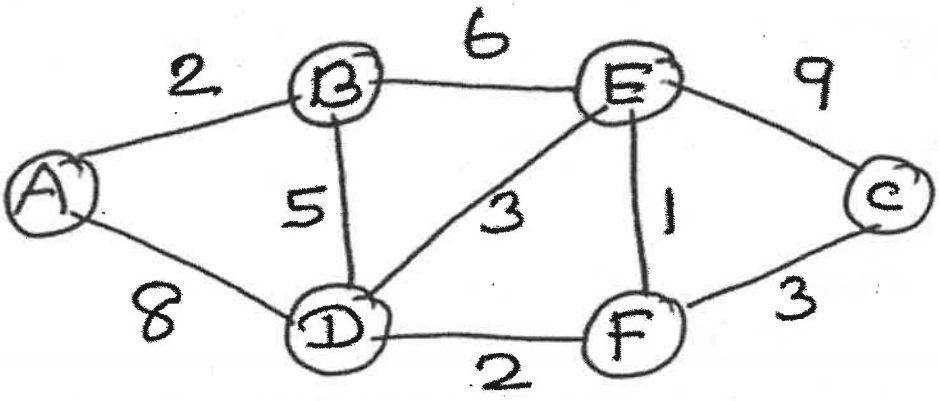
1. Attaching a compartment with the engine
2. Detaching a compartment from the engine

23. Sunil wants to rearrange his expression. He is familiar with the expression in which operator comes between variables. He wants to process the variables first then the operators. Help him to convert the expression given as follows: A+(B*C-(D/E^F)*G)*H. 10 3 3

24. Construct a binary search tree for the given data **45,39,56,12,34,78,32,10,89,54,67,81.** After constructing the tree, insert the elements **93** and **16** in the tree with proper pseudocode. 10 3 4

25. An IT company uses a 3-digit employee ID as a primary key. The employee ID of 10 employees are **100,201,303,405,602,600,702,801,927,829.** Consider a hash table of size 10 and insert the employee ID's in the hash table using division method. Apply linear probing method when 2 ID's map to the same location. 10 3 5

26. Apply Dijkstra's algorithm to find the shortest path cost from node A to C. 10 3 2



PART - C (1 × 15 = 15 Marks) Marks BL CO

Answer **any 1** Questions

27. Consider a data structure with the elements 22,32,44,51,65,71,80 which are stored in adjacent memory locations. The element 44 needs to be deleted first and after that you want to insert the data element 92 at the 4th position. Write the appropriate pseudo code to perform the aforementioned operations and provide a pictorial representation of it. 15 5 2

28. Harish plans to have a small library at his home. He has lot of books and he wants to assign a unique number to each book. This unique number helps him in identifying the position of the books on the bookshelf. The size of the bookshelf is 10. Help him to place the books in the bookshelf without any collision. The entries are **69,08,39,48,15,77,19** and discuss the collision avoidance strategies. 15 5 5

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Reg. No.														

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2024

Third Semester

21CSC201J – DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours | Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Answer **ALL** Questions

	Marks	BL	CO	PO
1. Which of the following case does not exist in complexity theory? (A) Best case (B) Worst case (C) Average case (D) Base case	1	1	1	1
2. How many times the following code prints the string "hello" for (i = 1; i<=50; i++) printf ("hello"); (A) 1 (B) 50 (C) Zero (D) 49	1	2	1	2
3. Elements in an array are accessed _____. (A) Randomly (B) Sequentially (C) Exponentially (D) Logarithmically	1	1	1	1
4. Which loop is guaranteed to execute at least one time (A) For (B) While (C) Do while (D) Both for and while	1	1	1	1
5. Linked list data structure offers considerable saving in (A) Computational time (B) Space utilization (C) Space utilization and computational time (D) Speed utilization	1	1	2	1
6. Which of the following is false about a double linked list? (A) Navigate in both the directions (B) It requires more space than a singly linked list (C) The insertion and deletion of a node take a bit longer (D) Implementing a double LL is easier than SLL	1	1	2	1
7. Which of the following application makes use of a circular linked list? (A) Undo operation in a text editor (B) Recursive function calls (C) Allocating CPU to resources (D) Implement hash tables	1	1	2	1

8. What is the functionality of the below code:
public void function (Node node)
{
if (size = = 0)
head = node;
	1	2	2	1

```
else
    Node temp, cur;
    for cur = head; (temp = cur.getNext( )! = null; cur = temp)        cur.setNext
(node);
        }
         size + +;
}
```

(A) Inserting a node at the beginning of the list
(B) Deleting a node at the beginning of the list
(C) Inserting a node at the end of the list
(D) Deleting a node at the end of the list

9. What does the following function check for? 1 3 3 2

```
#define max 10
typedef struct stack
{
 int top;
 int item [max];
} stack;
int function (stack *s)
{
 if (s→top = = -1)
    return 1;
else
    return 0;
}
```

(A) Full stack
(B) Invalid stack
(C) Empty stack
(D) Infinite stack

10. The postfix form of the expression $(A+B)*(C*D-E)*F/G$ 1 2 3 2

(A) $AB+CD*E-FG/**$
(B) $AB+CD*E-F**G/$
(C) $AB+CD*E-*F*G/$
(D) $AB+CDE*-*F*G/$

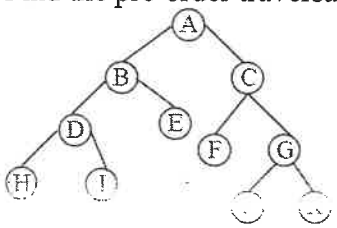
11. A linear list of elements in which deletion can be done from one end and insertion can take in place only at the other end is known as 1 1 3 1

(A) Queue
(B) Stack
(C) Tree
(D) Linked list

12. A data structure in which elements can be inserted or deleted at / from both ends but not in the middle is? 1 1 3 1

(A) Queue
(B) Circular queue
(C) Dequeue
(D) Priority queue

13. Find the pre-order traversal sequence. 1 2 4 2



(A) HIDEGFJKGCA
(B) HDIBEAFCJGK
(C) ABDHIECFGJK
(D) ABCDEFGHIJK

14. What is direct addressing? 1 2 4 1

(A) Distinct array position for every possible key
(B) Fewer array positions than keys
(C) Fewer keys than array positions
(D) Same array position for all keys

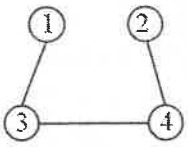
15. A binary search tree contains values 7, 8, 13, 26, 35, 40, 70. 75. Which one of the following is a valid post order sequence of the tree provided the pre-order sequence as 35, 13, 7, 8, 26, 70, 40 and 75? 1 1 4 2

(A) 7, 8, 26, 13, 75, 40, 70, 35 (B) 26, 13, 7, 8, 70, 75, 40, 35
(C) 7, 8, 13, 26, 35, 40, 70, 75 (D) 8, 7, 26, 13, 40, 75, 70, 35

16. What data organization method is used in hash tables? 1 1 4 1

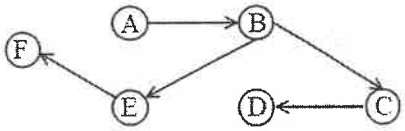
(A) Stack (B) Array
(C) Linked list (D) Queue

17. What would be the number of zero in the adjacency matrix of the given graph? 1 1 5 1



(A) 10 (B) 6
(C) 16 (D) 0

18. Which of the following is not a topological sorting of the given graph? 1 1 5 1

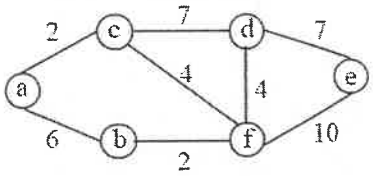


(A) ABCDEF (B) ABFEDC
(C) ABECFD (D) ABCDFE

19. The travelling salesman problem can be solved using. 1 1 5 1

(A) Spanning tree (B) A minimum spanning tree
(C) Bellman-Ford algorithm (D) DFS traversal

20. Consider the graph 1 1 5 1



What is the total cost of minimum spanning tree using Kruskals algorithm?

(A) 24 (B) 23
(C) 15 (D) 19

PART – B (5 × 8 = 40 Marks) Marks BL CO PO

Answer **ALL** Questions

21. a. How arrays can be created dynamically? Prove how it improves the performance of a program. 8 4 1 1

(OR)

b. Write a "C" program to store students information using structures. How effectively improved using pointers? 8 4 1 1

22. a. Consider an array A[1:n]. Given a position, write an algorithm to insert an element in the array. If the position is empty, the element is inserted easily. If the position is already occupied the element should be inserted with the minimum number of shifts. 8 4 2 2

(OR)

b. Describe the following 8 3 2 2

(i) Applications of list

(ii) Polynomial manipulation

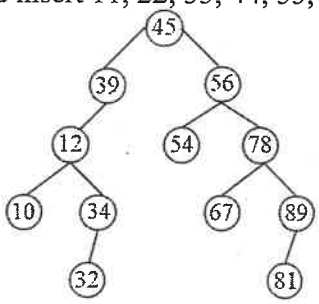
23. a. Develop and show the simulation using stack for evaluation of the following expression: $12+3*14-(5*16)+7$. 8 4 3 2

(OR)

b. Illustrate the steps to insert the following elements step by step in sequence into an empty AVL tree 15, 18, 20, 21, 28, 23, 30, 26. 8 4 3 2

24. a. Consider the binary search tree given below: 8 3 4 2

Show the result of in-order, pre-order and post-order traversal. Show the deletion of the root node and insert 11, 22, 33, 44, 55, 66 and 77 in the tree.



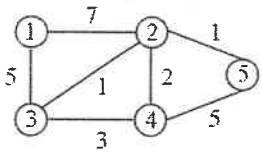
(OR)

b. Illustrate how different notations performed on a AVL tree with suitable example and routines. 8 3 4 2

25. a. Illustrate the algorithm to compute the shortest path using Dijkstra's algorithm. Validate the algorithm with suitable example. 8 4 5 2

(OR)

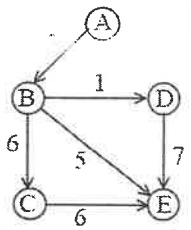
b. Illustrate Kruskal's algorithm to find the minimum spanning tree of a graph. Trace the algorithm for the following graph. 8 4 5 2



PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Question

	Marks	BL	CO	PO
26. Given input {4371, 1323, 6173, 4199, 4344, 9679, 1989} and a hash function $h(x) = x \bmod 10$, show the resulting (i) Open hash table (ii) Closed hash table using linear probing (iii) Closed hash table using quadratic probing	15	4	5	2
27. Apply an appropriate algorithm to find the shortest path from 'A' to every other node of A. For the given graph	15	4	5	2



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Reg. No.															

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JULY 2024

Third Semester

21CSC201J – DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

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(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Answer **ALL** Questions

	Marks	BL	CO	PO
1. Elements in an array are accessed (A) Randomly (B) Sequentially (C) Exponentially (D) Logarithmically	1	1	1	1

2. What will be the output of the following 'c' code?
struct student
{
char *c;
};
void main ( )
{
struct student *s;
s → c = "hello";
printf ("%.s", s→c);
}
(A) hello (B) Segmentation fault
(C) Run time error (D) Nothing
	1	2	1	1

3. The continue statement cannot be used with (A) for (B) while (C) do while (D) switch	1	2	1	1
4. Which of these best describes an array? (A) A data structure that shows a hierarchical behaviour (B) Container of objects of similar types (C) Arrays are immutable once initialized (D) Array is not a data structure	1	1	1	1
5. Which of the following is an example of linear DS? (A) Tree (B) Linked list (C) Queue (D) Linked list and queue	1	2	1	1
6. In linked list implementation a node carries information regarding. (A) Data (B) Link (C) Data and link (D) Node	1	1	2	1

7. Which of the following is false about a circular linked list? 1 1 2 1

(A) Every node has a successor
(B) Time complexity of inserting a new node at the head of the list is O(1)
(C) Time complexity for deleting the last node is O(n)
(D) We can traverse the whole circular linked list starting from any point

8. Which of the following application makes use of a circular linked list? 1 2 2 1

(A) Undo operation in a text editor
(B) Recursive function calls
(C) Allocating CPU to resources
(D) Implement hash tables

9. The data structure required to check whether an expression contains balanced parenthesis is? 1 1 3 1

(A) Stack
(B) Queue
(C) Array
(D) Tree

10. The prefix form of $A - B / (C * D \wedge E)$ is? 1 1 3 2

(A) $- / * \wedge ACBDE$
(B) $-ABCD * \wedge DE$
(C) $-A / B * C \wedge DE$
(D) $-A / BC * \wedge DE$

11. A queue follows 1 1 3 2

(A) FIFO (First In First Out) principle
(B) LIFO (Last In Last Out) principle
(C) Ordered array
(D) Linear tree

12. A normal queue, if implemented using an array of size max_size, gets full when? 1 2 3 1

(A) Rear = max_size –1
(B) Front = (rear –1) mod max_size
(C) Front = rear + 1
(D) Rear = front

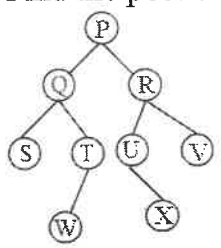
13. What is complete binary tree? 1 1 4 1

(A) Each node has exactly zero or two children
(B) Binary tree, which is completely filled with possible exception of the bottom level which is filled from right to left
(C) A binary tree, which is completely filled with the possible exception of the bottom level, which is filled from left to right
(D) A tree in which all nodes have degree Z

14. In a _____, the value of the left node must be smaller than the parent node and value of the right node must be greater then the parent node. 1 1 4 1

(A) Full binary tree
(B) Complete binary tree
(C) Binary search tree
(D) Balanced binary tree

15. Find the post order traversal of the binary tree shown below? 1 2 4 3

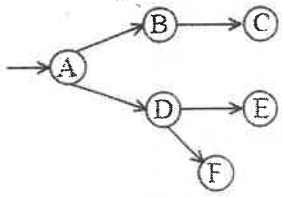


(A) PQRSTUVWX (B) WRSQPVTUX
(C) SWTQXUVRP (D) STWUXVQRP

16. What is a hash function? 1 1 4 1
(A) A function has allocated memory to keys (B) A function that computes the location of the key in the array
(C) A function that creates an array (D) A function that computes the location of the values in the array

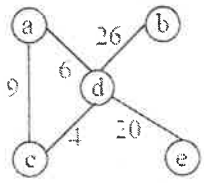
17. For the adjacency matrix of a directed graph, the row sum is the _____ degree and the column sum is the _____ degree. 1 2 5 3
(A) In, out (B) Out, in
(C) In, total (D) Total, out

18. What sequence would the BFS traversal of the given graph yield? 1 2 5 3



(A) AFDBCE (B) CABAFED
(C) ABDCEF (D) EFDCBA

19. Consider the graph below, which of the following are the edges in the MST of the given graph? 1 1 5 1



(A) (a-c) (c-d) (d-b) (d-b) (B) (c-a) (a-d) (d-b) (d-e)
(C) (a-d) (d-c) (d-b) (d-e) (D) (c-a) (a-d) (d-c) (d-b) (d-e)

20. Which of the following is not the algorithm to find the minimum spanning tree of the given graph? 1 1 5 3
(A) Boruvkn's algorithm (B) Prims algorithm
(C) Kruskal's algorithm (D) Bellman-Ford algorithm

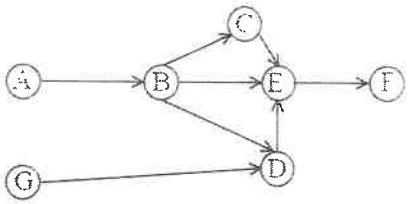
PART – B (5 × 8 = 40 Marks) Marks BL CO PO
Answer **ALL** Questions

21. a. Consider an array A[1:n]. Given a position, write an algorithm to insert an element in the array. If the position is empty, the element is inserted easily. If the position is already occupied, the element should be inserted with the minimum number of shifts. 8 4 1 1

(OR)

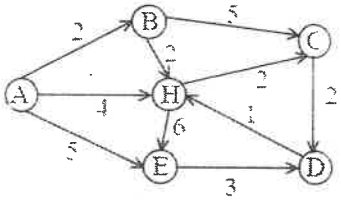
b. Write a 'C' program to store information of employees using user defined structures and pointers. 8 4 1 1

22. a.	Write an algorithm to demonstrate a polynomial using a linked list for addition of two polynomials.	8	3	2	2
	(OR)				
b.	How an element can be deleted from a doubly linked list? Explain with suitable routine and diagram.	8	4	2	2
23. a.	Write an algorithm to convert an infix expression to a postfix expression. Trace the algorithm to convert the infix expression $(a+b)*c/d+e/f$ to a postfix expression.	8	4	3	2
	(OR)				
b.	Give algorithm for PUSH and POP operations on stack using a linked list with suitable pictorial representations.	8	3	3	2
24. a.	Illustrate how different rotations performed on a AVL tree with suitable example and routines.	8	3	4	2
	(OR)				
b.	Create a B+ tree of order-5 for the following data arriving in sequence. 90, 27, 7, 9, 18, 21, 3, 4, 16, 11, 21, 72	8	4	4	2
25. a.	How DFS be performed on an undirected graph. Illustrate with an example.	8	3	5	2
	(OR)				
b.	Consider the given directed acyclic graph D. Sort the nodes D by applying topological sort on 'D'.	8	4	5	2

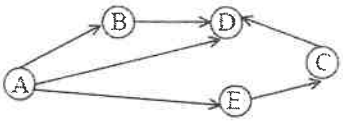


PART – C (1 × 15 = 15 Marks)
Answer **ANY ONE** Question

		Marks	BL	CO	PO
26.	Using Dijkstra's algorithm find the shortest path from the source node A.	15	4	5	2



27.i.	When do you perform rehashing? Illustrate with example.	8	4	5	2
ii.	From the given in what order the vertices visited using DFS and BFS starting from vertex A? Explain with algorithm.	7	4	5	2



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Reg. No.															

B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Third Semester

21CSC201J - DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year2021-2022 to 2023-2024)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours Max. Marks: 75

PART - A (20 x 1 = 20 Marks) **Marks BL CO PO**

Answer ALL Questions

1. _____are the set of definitions that allow us to use the functions while hiding the implementation 1 1 1 2

A) Abstract Data Types B) Algorithms

C) Referential Structures D) Datatypes

2. What is the output if it is executed on a 32 bit processor? 1 1 1 2

```
void main()
{
int *p;
p = (int *)malloc(20);
printf("%d", sizeof(p));
}
```

A) 2 B) 4

C) 8 D) Compiler dependent

3. Which of the following functions allocates multiple blocks of memory, each block of the same size? 1 1 1 3

A) malloc() B) realloc()

C) calloc() D) free()

4. We can certainly deduce the _____ scenario of an algorithm from its asymptotic analysis. 1 1 1 12

A) Best case B) Best, worst and average cases

C) Worst case D) Average case

5. Identify the correct option for creating a node in linked list at first time 1 1 2 3

A) ptr = (struct node) malloc (sizeof(struct node *));

B) ptr = (struct node *)calloc(sizeof(struct node *));

C) ptr = (struct node *)malloc(sizeof(struct node));

D) ptr = (struct node *) calloc (size(struct node));

6. The matrix contains m rows and n columns. The matrix is called Sparse Matrix if _____ 1 1 2 3

A) Total number of Zero elements > (m*n)/2

B) Total number of Zero elements = m + n

C) Total number of Zero elements = m/n

D) Total number of Zero elements = m-n

7. Find out the correct option when adding a new node into the end of list 1 2 2 2

A) Head is made to point to the new node, New node points to the previously first element

B) Last node now points to the new node, New node points to NULL

C) Previous node now points to the new node, New node points to the next node

D) Last node now points to the new node, New node points to the previously first element

8. Which of the following is a practical example of a doubly linked list 1 1 2 3

A) A browser cookie file
B) A quest in a game that lets users retry stages
C) A game in which the player runs forward
D) A first-in-first out scheduling system

9. A stack data structure cannot be used for 1 1 3 2

A) Implementation of Recursive Function
B) Allocation Resources and Scheduling
C) Reversing string
D) Evaluation of string in postfix form

10. The five following names Allen, Brindha, Chitra, Divya and Elon are pushed in a stack, one after another starting from Allen. The stack is popped four times and each name is inserted in a Queue. Then two names are deleted from the Queue and pushed back on the stack. Now one name is popped from stack. The popped name is _____. 1 3 3 2

A) Allen
B) Brindha
C) Chitra
D) Divya

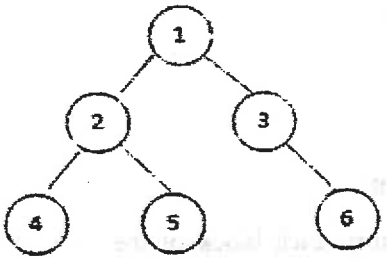
11. If the elements "A", "B", "C" and "D" are placed in a queue and are deleted one at a time, in what order will they be removed? 1 1 3 2

A) ABCD
B) DCBA
C) DCAB
D) ABDC

12. Identify the Postfix form of following expression. D + (E * F^G) 1 1 3 12

A) DFG^E * +
B) DEFG^* +
C) GFED^ *+
D) GF^E* D+

13. Which one is the correct In-order traversal for the given tree 1 1 4 3



A) 1->2->3->4->5->6
B) 4->2->5->1->3->6
C) 4->5->2->6->3->1
D) 4->2->5->3->6->1

14. A B-tree of order 4 and of height 3 will have a maximum of _____ keys. 1 1 4 2

A) 255
B) 63
C) 127
D) 188

15. What is a hash table? 1 1 4 3

A) A structure that maps values to keys
B) A structure used for storage
C) A structure that maps keys to values
D) A structure used to implement stack and queue

16. If several elements are competing for the same bucket in the hash table, what is it called? 1 1 4 3

A) Diffusion
B) Replication
C) Collision
D) Duplication

17. Which of the following statements is/are TRUE for an undirected graph? 1 1 5 12

P: The number of odd-degree vertices is even

Q: Sum of degrees of all vertices is even

A) P Only
B) Q only
C) Both P and Q
D) Neither P nor Q

18. Consider an undirected unweighted graph G. Let a breadth-first traversal of G be done starting from a node r. Let d(r, u) and d(r, v) be the lengths of the shortest paths from r to u and v respectively, in G. If u is visited before v during the breadth-first traversal, which of the following statements is correct? 1 1 5 12

A) d(r, u) < d (r, v)
B) d(r, u) > d(r, v)
C) d(r, u) <= d (r, v)
D) d(r, u) =d(r, v)

19. The efficiency of open addressing techniques 1 1 5 3

A) Depends on the size of the table and number of elements in the table
B) Depends on the size of the table
C) Depends on total number of elements in the table
D) Depends on the hashing technique

20. Which of the following algorithms is commonly used to find a minimum spanning tree in a connected undirected graph? 1 1 5 3

A) Dijkstra's algorithm
B) Kruskal's algorithm
C) Bellman-Ford algorithm
D) Floyal-Warshall algorithm

PART - B (5 x 8 = 40 Marks)
Answer **ALL** Questions

Marks BL CO PO

21 a. Calculate the time complexity for the following code using Tabular method. Specify the asymptotic time complexity (worst case) for the given algorithm. 8 1 1 2

```
void calculate (int arr[], int n)
{
int sum = 0;
for (int i = 0; i < n; i++)
{
for (int j = i; j < n; j++)
{
for (int k = 0; k < 10; k++)
{ sum += arr[j];
} } } printf("%d\n", sum); }
```

(OR)

b. Discuss the concept of self-referential structures in the C programming language. How are they typically used in the implementation of dynamic data structures like linked lists? Illustrate with an example, including code snippets, to demonstrate how a self-referential structure can be used to implement a basic linked list. 8 1 1 1

22 a. Write an algorithm that takes in a two-dimensional array and find the maximum of all the row sums. A row sum is simply the sum of each number on that row? What are the limitations of array data structure 8 3 2 1

(OR)

b. Consider a goods train, which initially has an engine as a header. Later compartments are linked with engines one by one in series. The compartments can be attached and detached with an engine or in between the compartments. Implement a data structure for the following operations in the goods train 1. Attaching a compartment in between the compartments 2. Detaching a compartment in between the compartments. 8 3 2 2

23 a. Write the pseudocode to convert the expression given below into its corresponding postfix expression and then evaluate it. 8 3 3 12

10 + ((7 – 5) + 10)/2

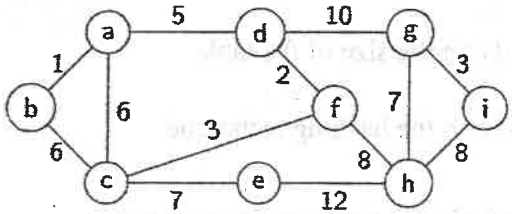
(OR)

b. Write the pseudocode to insert an element in the linked queue and delete an element from the linked queue with an example. 8 3 3 12

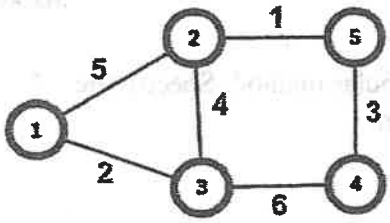
24 a. Mythili wants to constructs a tree for the given data **15, 10, 8, 12, 28, 20, 47, 37, 64 and 2** in such a way that the value of the nodes in right sub tree should be greater than the left sub tree. Then she felt that the leaves in the constructed tree will not be in same order, so she decided to remove the data 8, 28 and 64. Construct the tree as per Mythili's wish. 8 3 4 1

(OR)

b. Construct the minimum spanning tree using suitable algorithm and justify its efficiency 8 3 4 2



25 a. Draw the possible spanning trees for the given Graph G. Identify the minimum spanning tree among the spanning trees drawn. Write the pseudocode for generating spanning trees. 8 3 5 2



(OR)

b. Harish plans to have a small library at his home. He has lot of books and he wants to assign a unique number to each book. This unique number helps him in identifying the position of the books on the bookshelf. The size of the bookshelf is 10. Help him in place the books in the bookshelf without any collision. The entries are **69, 08, 39, 48, 15, 77, 19.** Compare the state of the structure by using two collision avoidance strategies. 8 3 5 2

PART - C (1 x 15 = 15 Marks) **Marks BL CO PO**

Answer **ANY ONE** Question

26. A company approaches you to develop a customer support ticket system, they might add new tickets and remove tickets, from anywhere. Suggest a suitable abstarct data type. Implement using appropriate structure and provide pseudocode for 15 3 2 3

1. Insert ticket at the end
2. Remove ticket from the beginning
3. Insert at the beginning
4. Delete the last ticket

Justify why you suggested the ADT

27. Construct two trees for the elements 11, 22, 33, 44, 55, 66, 77, 88 15 3 4 3

Property 1: Left subtree should have elements smaller than the element at parent and elements of right subtree should be greater than that of the parent.

Peroperty 2: Difference in the height of the left subtree and right subtree should not greater than 1.

First tree should satisfy property1 and the second tree should satisfy both the properties.

******

(OR)

b. How an element can be deleted from a doubly linked list? Explain with suitable routine and diagram. 8 4 2 2

23. a. Develop and show the simulation using stack for evaluation of the following expression: $12+3*14-(5*16)+7$. 8 4 3 2

(OR)

b. Give algorithm for PUSH and POP operations on stack using a linked list with suitable pictorial representations. 8 4 3 2

24. a. Construct the AVL tree for the given data: 8 3 4 2

22, 25, 31, 10, 5, 15, 29, 19, 16, 11, 3, 4, 8

(OR)

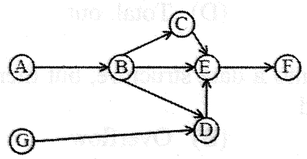
b. Construct the binary search tree for the following data: 8 4 4 2

50, 30, 10, 90, 100, 40, 60, 20, 110, 5

25. a. How DFS be performed on an undirected graph. Illustrate with an example. 8 3 5 2

(OR)

b. Consider the given directed acyclic graph D. Sort the nodes D by applying topological sort on 'D'. 8 4 5 2

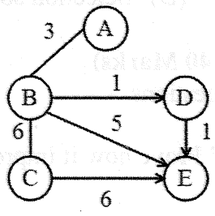


PART – C (1 × 15 = 15 Marks) Marks BL CO PO

Answer **ANY ONE** Question

26. The keys 2, 4, 13, 5, 26, 7, 19, 21, 9 are to be sorted. Use any two data structure to store these keys, Discuss the advantages and disadvantages of each of the proposed structure. 15 4 5 2

27. Apply an appropriate algorithm to find the shortest path from 'A' to every other node of A. For the given graph. 15 3 2 2



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Reg. No.														

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Third Semester

21CSC201J – DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year 2021-2022 to 2024-2025)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

PART – A (20 × 1 = 20Marks) Marks BL CO PO

Answer **ALL** Questions

1. Elements in an array are accessed 1 1 1 1

(A) Randomly (B) Sequentially

(C) Exponentially (D) Logarithmically

2. What will be the output of the following 'c' code? 1 2 1 1

```
struct student
 {
  char *c;
 };
void main ( )
 {
  struct student *s;
  s → c = "hello";
  printf ("%s", s→c);
 }
```

(A) hello (B) Segmentation fault

(C) Run time error (D) Nothing

3. Consider a hash table with 98 slots. The has function is h(k) = k mod 9. The collisions are resolved by chaining. The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. The maximum, minimum and average chain lengths in the hash table, respectively are 1 5 5 2

(A) 3, 0 and 1 (B) 3, 3 and 3

(C) 4, 0 and 1 (D) 3, 0 and 2

4. A priority queue is implements as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is: 10, 8, 5, 3, 2. Two new elements 1 and 7 are inserted into the heap in that order. The level-order traversal of the heap after the insertion of the element is 1 3 3 2

(A) 10, 8, 7, 3, 2, 1, 5 (B) 10, 8, 7, 2, 3, 1, 5

(C) 10, 8, 7, 1, 2, 3, 5 (D) 10, 8, 7, 5, 3, 2, 1

5. Which of the following is an example of linear DS? 1 2 1 1

(A) Tree (B) Linked list

(C) Queue (D) Linked list and queue

6. Let G be a graph with 'n' vertices and 'm' edges. What is the tightest upper bound on the running time on Depth First Search of G? Assume that the graph is represented using adjacency matrix. 1 3 5 1
   (A) O(n) (B) O(n+m)
   (C) O(n^2) (D) O(mn)

7. Which of the following is false about a circular linked list? 1 1 2 1
   (A) Every node has a successor (B) Time complexity of inserting a new node at the head of the list is O(1)
   (C) Time complexity for deleting the last node is O(n) (D) We can traverse the whole circular linked list starting from any point

8. The following postfix expression with single digit operands is evaluated using a stack: 1 3 3 1
   8 2 3 ^ / 2 3 * + 5 1 * -
   Note that ^ is the exponentiation operator. The top two elements of the stack after the * is evaluated are
   (A) 6, 1 (B) 5, 7
   (C) 3, 2 (D) 1, 5

9. What is the maximum height of any AVL-tree with 7 nodes? Assume that the height of a tree with a single node is 0. 1 2 4 1
   (A) 2 (B) 3
   (C) 4 (D) 5

10. Which one of the following in place sorting algorithms needs the minimum number of swaps? 1 1 3 2
    (A) Quick sort (B) Insertion sort
    (C) Selection sort (D) Heap sort

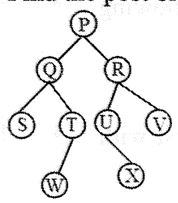
11. A queue follows 1 1 3 2
    (A) FIFO (First In First Out) principle (B) LIFO (Last In Last Out) principle
    (C) Ordered array (D) Linear tree

12. The best data structure to check whether an arithmetic expression has balanced parentheses is 1 1 3 1
    (A) Queue (B) Stack
    (C) Linked list (D) Tree

13. What is complete binary tree? 1 1 4 1
    (A) Each node has exactly zero or two children
    (B) Binary tree, which is completely filled with possible exception of the bottom level which is filled from right to left
    (C) A binary tree, which is completely filled with the possible exception of the bottom level, which is filled from left to right
    (D) A tree in which all nodes have degree Z

14. Level order traversal of a rooted tree can be done by starting from the root and performing? 1 1 4 1
    (A) Pre order traversal (B) In order traversal
    (C) Depth first search (D) Breadth first search

15. Find the post order traversal of the binary tree shown below? 1 2 4 3



    (A) PQRSTUVWX (B) WRSQPVTUX
    (C) SWTQXUVRP (D) STWUXVQRP

16. What is a hash function? 1 1 4 1
    (A) A function has allocated memory to keys (B) A function that computes the location of the key in the array
    (C) A function that creates an array (D) A function that computes the location of the values in the array

17. For the adjacency matrix of a directed graph, the row sum is the _____ degree and the column sum is the _____ degree. 1 2 5 3
    (A) In, out (B) Out, in
    (C) In, total (D) Total, out

18. When new data are to be inserted into a data structure, but there is not available space, this situation is usually called 1 1 5 1
    (A) Underflow (B) Overflow
    (C) Saturated (D) Houseful

19. In an unweighted, undirected connected graph, the shortest path from a node S to every other node is computed most efficiently, in term of time complexity by 1 3 5 1
    (A) Dijkstra's algorithm starting from S (B) Warshall's algorithm
    (C) Performing a DFS starting from S (D) Performing a BFS starting from S

20. Which of the following sorting algorithm has the lowest worst-case complexity? 1 2 5 1
    (A) Merge sort (B) Quick sort
    (C) Bubble sort (D) Selection sort

PART – B (5 × 8 = 40 Marks) Marks BL CO PO
Answer **ALL** Questions

21. a. How arrays can be created dynamically? Prove how it improves the performance of a program. 8 4 1 1

(OR)

    b. Write a 'C' program to store students information using structures. How effectively improved using pointers? 8 4 1 1

22. a. Write an algorithm to demonstrate a polynomial using a linked list for addition of two polynomials. 8 3 2 2

Reg. No.															

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Third Semester

21CSC201J - DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year 2021-2022 to 2024-2025)

Note:
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(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours | Max. Marks: 75

PART - A (20 x 1 = 20 Marks)
Answer **ALL** Questions

		Marks	BL	CO	PO
1.	Which of the following is not a type of pointer in C? A) NULL pointer B) Void pointer C) Generic pointer D) String pointer	1	1	1	1
2.	Which of the following is a self-referential structure? A) struct node { int data; struct node* next; } B) struct node { int data; } C) struct node { int data; char next; } D) struct node { int data; double next; }	1	1	1	1
3.	Examine the correct way of dynamically allocating memory for an integer array of size 10 in C? A) int *arr = (int*) malloc(10) B) int *arr = (int*) malloc(10 * sizeof(int)) C) int *arr = (int*) malloc(sizeof(int)) D) int *arr = (int*) malloc(10 * sizeof(char))	1	2	1	1
4.	Which of the following describes the time complexity O(n)? A) Constant time B) Logarithmic time C) Linear time D) Quadratic time	1	1	1	1
5.	When implementing a list ADT using an array, what is the primary drawback when inserting an element at the beginning of the list? A) Requires shifting all elements to the right B) Requires extra space for the new element C) Cannot insert at the beginning D) Elements are not contiguous in memory	1	1	2	1
6.	Suppose the contents of an array A is {1, null, null, null, null, 10}; What would be the size of the structure to store the elements using normal array and sparse array? A) 6 and 6 B) 6 and 2 C) 2 and 6 D) 2 and 2	1	2	2	1
7.	In Josephus problem, when 7 people stand in the circle waiting to be executed and every 3rd person is eliminated in each round. Identify which of the following is the correct Elimination Sequence? A) 3, 6, 2, 7, 5, 1 B) 3, 2, 7, 6, 5, 1 C) 3, 5, 2, 7, 6, 1 D) 3, 6, 1, 7, 5, 2	1	2	2	1
8.	How does the cursor-based list differ from the standard array implementation in terms of memory management? A) Cursor-based lists are dynamically allocated B) Cursor-based lists use a fixed-size array to manage elements and their "pointers" C) Cursor-based lists do not manage memory D) Cursor-based lists use pointers like linked lists	1	2	2	1

9. What is the result of the given operation Top (Push (S, X))? 1 2 3 2

A) X
B) Null
C) S
D) -1

10. The postfix form of the expression (A+ B)*(C*D– E)*F / G is 1 2 3 2

A) AB+ CD*E – FG /**
B) AB + CD* E – F **G /
C) AB + CD* E – *F *G /
D) AB + CDE * – * F *G /

11. Queues serve major role in 1 2 3 1

A) Simulation of recursion
B) Simulation of arbitrary linked list
C) Simulation of limited resource allocation
D) Simulation of heap sort

12. The Circular Queue will be full only when 1 2 3 1

A) Front max-1 and Rear max-1
B) Front=0 and Rear=Max-1
C) Front=Max-1 and Rear=0
D) Front=0 and Rear=0

13. Given a full binary tree with n leaves, the total number of nodes in the tree is _____ 1 2 4 1

A) n
B) 2n - 1
C) 2n
D) 2n + 1

14. When do you need to perform a rotation in an AVL tree to maintain balance? 1 1 4 1

A) After every insertion or deletion
B) Only after insertion
C) Only after deletion
D) Only when the tree becomes unbalanced after a insertion or deletion

15. Which of the following is not a common hash function? 1 1 4 1

A) Division method
B) Multiplication method
C) Folding method
D) Expansion method

16. What is the main drawback of using chaining for collision resolution? 1 1 4 1

A) It requires more memory
B) It is difficult to implement
C) It can lead to unbalanced chains and increased search time
D) It requires constant rehashing

17. The data structure commonly used to perform Depth-First Search (DFS) in a graph is 1 2 5 1

A) Queue
B) Stack
C) Heap
D) Priority Queue

18. Which of the following is a type of graph that has no cycles? 1 1 5 1

A) Complete graph
B) Directed graph
C) Tree
D) Weighted graph

19. Topological sorting for Directed Acyclic Graph, order the vertices in _____ order 1 2 5 1

A) Ascending
B) Descending
C) Linear
D) Non Linear

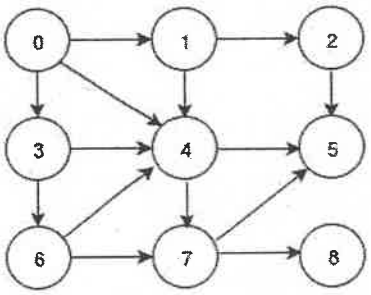
20. Which of the following is false about Prim's algorithm? 1 1 5 1

A) It is a greedy algorithm
B) It constructs MST by selecting edges in increasing order of their weights
C) It never accepts cycles in the MST
D) It can be implemented using the Fibonacci heap

PART - B (5 x 8 = 40 Marks)

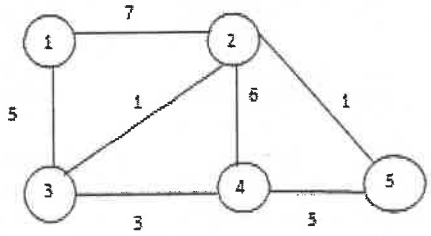
Answer **ALL** the Questions

		Marks	BL	CO	PO
21 a.	Explain passing structures to a function with suitable examples for the following, i) Passing individual members. ii) Passing entire structure.	8	1	1	1
	(OR)				
b.	Illustrate briefly on Big oh Notation, Omega Notation and Theta Notations. Depict the same graphically and explain.	8	1	1	1
22 a.	Illustrate an algorithm by assuming your own data to i) Count the number of nodes in a given singly linked list. ii) Searching the element from singly linked list.	8	2	2	2
	(OR)				
b.	Recommend an algorithm to add two polynomials represented by linked representation. Apply the function for the following input. $A = 3x^2 + 2x + 1$, $B = 8x^2 + 3x + 10$	8	2	2	2
23 a.	Show the procedure to convert the infix expression to postfix expression and steps involved in evaluating the postfix expression. Convert the expression (A + B / C * (D + E) – F) to postfix form	8	2	3	2
	(OR)				
b.	Illustrate the Dequeue and its types? Write the algorithm for the Insertion operation at front end and Deletion operation at rear end in a Dequeue.	8	2	3	2
24 a.	What are AVL trees? Illustrate the different rotations which takes place while inserting the following elements into an empty AVL tree 63, 9, 19, 27, 18, 108, 99, 81.	8	2	4	3
	(OR)				
b.	Given input {4371 1323, 6173, 4199,4344,9679,1989} and a hash function h(x) =x mod10. Prepare the following (i) Open addressing hash table using linear probing. (ii) Open addressing hah table using quadratic probing	8	2	4	3
25 a.	Given a graph (G), In what order are the vertices visited using BFS and DFS starting from vertex 0? Illustrate with required steps.	8	2	5	3



(OR)

b.	Illustrate Prim's algorithm to find the minimum spanning tree of the given graph.	8	2	5	3



PART - C (1 x 15 = 15 Marks)
Answer **ANY ONE** Questions

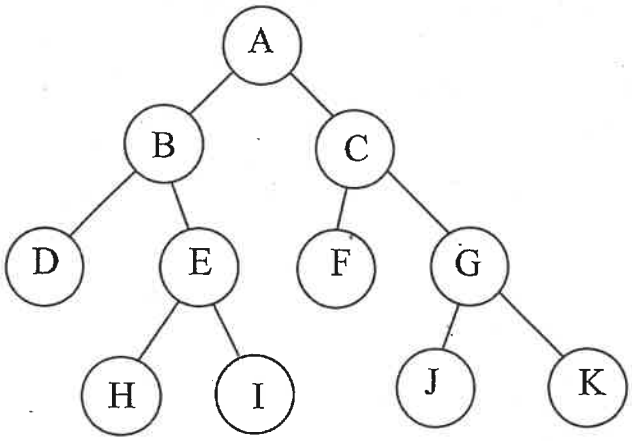
Marks BL CO PO

26 (i) A circular queue of size 5, has 3 elements 10, 20 and 40 with the index of Front=2 and Rear=4. 15 3 2 2

1. Insert 50 and 60.
2. Insert 30.
3. Delete two elements
4. Insert 70, 80 & 90.

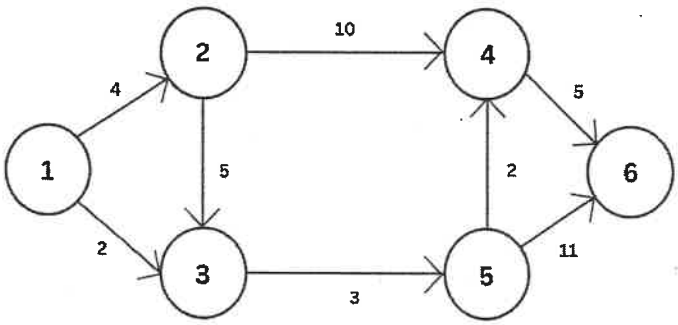
Assess the sequence of steps with necessary diagrams and the location of Front & Rear. (8 Marks)

(ii) Apply the traversal algorithms and list the order in which the nodes would be traversed. List scenarios suitable for each type of traversal. (7 Marks)



27 (i) Depict the construction of B Tree whose order is 5 to insert the following keys 1, 12, 8, 2, 25, 5, 14, 28, 17, 7, 52, 16, 48, 68, 3, 26, 29, 53, 55, 45. Enumerate the rules for deletion and delete the key 8 followed by 55. (8 Marks) 15 3 4 2

(ii) Consider the Given Graph, find Minimum Spanning Tree by considering the node 1 as the root of the tree. (7 Marks)



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Reg. No.															

B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, JUNE 2025

Third Semester

21CSC201J - DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year 2021-2022 TO 2024-2025)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours | Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer **ALL** Questions

		Marks	BL	CO	PO
1.	How do you access a member of a structure using a pointer in C? A) ptr.member B) ptr->member C) *ptr.member D) ptr&.member	1	1	1	2

2.	What is the output of the following code?
	int x = 10;
	int *p = &x;
	printf("%d", *p);
	A) Address of x B) 10
	C) Garbage value D) Compilation error
		1	2	1	1

3.	Which of the following statements about matrix multiplication is true? A) Matrix multiplication is commutative B) Matrix multiplication is associative C) Matrix multiplication is distributive over addition D) Both b and c	1	2	1	1
4.	What is the best time complexity of a linear search? A) O(1) B) O(n) C) O(log n) D) $O(n^2)$	1	1	1	2
5.	In a singly linked list, how is an element inserted at the beginning of the list? A) By creating a new node and updating the head pointer B) By shifting all elements one position to the right C) By swapping the first two nodes D) By appending the element at the end	1	3	2	2
6.	What is the disadvantage of inserting an element at the beginning in an array-based implementation of List ADT? A) Requires extra space B) Takes constant time C) Requires shifting all elements D) Cannot be done	1	2	2	3
7.	How is the end of the list detected in a circular singly linked list? A) When the next pointer is NULL B) When the next pointer points to the head C) When the previous pointer is NULL D) When the previous pointer points to the tail	1	2	2	2

8. In the Josephus problem, how is the next person to be eliminated chosen in a circular linked list? 1 3 2 3

A) Randomly
B) By a fixed number of steps
C) By a linear search
D) By sorting the list

9. What is the time complexity of the push operation in a stack implemented using an array? 1 1 3 2

A) O(1)
B) O(n)
C) O(log n)
D) O(n log n)

10. What is the postfix expression for the infix expression (A + B) * (C + D)? 1 3 3 2

A) AB + CD + *
B) AB + C + D *
C) AB + CD * +
D) ABCD + * +

11. How many moves are required to solve the Tower of Hanoi problem with 4 disks? 1 3 3 2

A) 16
B) 15
C) 14
D) 24

12. In a circular queue implemented using an array, how do you check if the queue is full? 1 2 3 3

A) front == rear
B) (rear + 1) % size == front
C) front == size
D) rear == size

13. In a binary tree, the maximum number of nodes at level L is: 1 1 4 3

A) 2^L
B) $2^{(L-1)}$
C) $2^{(L+1)}$
D) L^2

14. A complete binary tree is defined as: 1 1 4 2

A) A binary tree in which all levels are completely filled.
B) A binary tree in which all levels except possibly the last are completely filled.
C) A binary tree with nodes having at most two children.
D) A binary tree in which each node has exactly two children.

15. In an AVL tree, which rotation is applied when the tree becomes left-heavy due to the insertion of a new node in the left subtree of the left child? 1 3 4 2

A) Right rotation
B) Left rotation
C) Left-right rotation
D) Right-left rotation

16. What is the maximum height of a Red-Black tree with n nodes? 1 2 4 3

A) 2*log(n+1)
B) 2*log(n)
C) log(n+1)
D) log(n)

17. Quadratic probing is primarily used to: 1 2 5 3

A) Avoid clustering.
B) Handle large datasets.
C) It is only applicable to undirected graphs.
D) All edges must point in the same direction.

18. In a topological sorting of a graph, which of the following is true? 1 2 5 3

A) The graph must be cyclic.
B) It is possible to have multiple valid orderings.
C) It is only applicable to undirected graphs.
D) All edges must point in the same direction.

19. Which of the following data structures is commonly used to implement Kruskal's algorithm? 1 1 5 2

A) Stack
B) Queue
C) Disjoint Set (Union-Find)
D) Priority Queue

20. In Dijkstra's algorithm, which data structure is commonly used to select the next vertex with the minimum distance? 1 1 5 3

A) Stack
B) Queue
C) Priority Queue (Min-Heap)
D) Hash Table

PART - B (5 x 8 = 40 Marks)

Answer **ALL** the Questions

		Marks	BL	CO	PO
21 a.	Define and explain the differences between primitive data types and structures in programming. Provide an example in C demonstrating the declaration and usage of both.	8	2	1	2
	(OR)				
b.	Discuss dynamic memory allocation in C and illustrate how to dynamically allocate memory for a 2D matrix. Also, provide a simple example of matrix multiplication.	8	2	1	3
22 a.	Define and differentiate between Singly Linked List, Doubly Linked List, and Circular Linked List. Provide a scenario where each type is preferred.	8	2	2	3
	(OR)				
b.	Explain the Josephus Problem. Describe how a Circular Linked List can be used to solve this problem. Write the algorithm for the same.	8	2	2	2
23 a.	Describe the priority queue and its implementation using a linked list. How does it differ from a normal queue?	8	2	3	3
	(OR)				
b.	What is a circular queue? How is it different from a linear queue? Illustrate with an example.	8	2	3	3
24 a.	Describe the different ways to represent a binary tree in memory. Compare the advantages and disadvantages of using a linked representation versus an array representation.	8	2	4	3
	(OR)				
b.	Explain the properties of a Red-Black Tree. How do these properties ensure that the tree remains balanced? Illustrate with an example.	8	2	4	3
25 a.	Describe Depth-First Search (DFS) and Breadth-First Search (BFS) graph traversal techniques. Compare their applications.	8	2	5	2
	(OR)				
b.	Explain the process of topological sorting of a Directed Acyclic Graph (DAG). What are its applications?	8	2	5	3

PART - C (1 x 15 = 15 Marks)

Answer **ANY ONE** Questions

		Marks	BL	CO	PO
26	Discuss the importance of mathematical notations in analyzing algorithms. Explain asymptotic notations Big O, Omega, and Theta with examples, and describe how they are used to express the efficiency of algorithms.	15	2	1	3
27	Explain the concept of open addressing in hashing. Discuss the various probing techniques used in open addressing, including linear probing, quadratic probing, and double hashing. Evaluate their performance in terms of clustering and efficiency.	15	2	5	3

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23. a. Write an algorithm to convert an infix expression to a postfix expression. Trace the algorithm to convert the infix expression a+b ∗ c/d+e/f to a postfix expression. 8 4 3 3

(OR)

b. Give algorithms for push and pop operations on stack using a linked list with suitable pictorial representations. 8 3 3 2

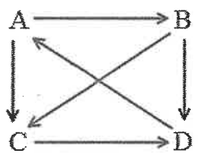
24. a. Write appropriate algorithm for inserting and deleting a node in a binary search tree. 8 4 3 3

(OR)

b.i. Create a B+tree of order – 5 for the following data arriving in sequence 90, 27, 7, 9, 18, 21, 3, 4, 16, 11, 21, 72 4 4 4 2

ii. Compare BST with AVL tree. 4 3 4 2

25. a. Consider the graph given below show its adjacency list and adjacency matrix in the memory. 8 5 5 2



(OR)

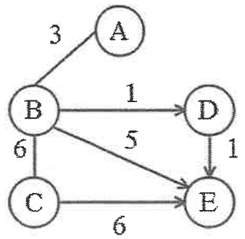
b. Illustrate Kruskal's algorithm to find the minimum spanning tree of a graph. 8 4 5 2

PART – C (1 × 15 = 15 Marks) Marks BL CO PO

Answer **ANY ONE** Questions

26. Consider a hash table with 9 slots. The hash function is h(k) = k mod 9. The following keys are inserted in order 5, 28, 19, 15, 20, 33, 12, 17, 10. Draw the contents of the hash table when the collisions are solved by
   i. Chaining
   ii. Linear probing
   iii. Double hashing. The second hash function $h_2(x) = 7 - (x\%7)$.

27. Apply an appropriate algorithm to find the shortest path from 'A' to every other node of A. For the given graph.



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Reg. No.														

B.Tech. DEGREE EXAMINATION, MAY 2023

Third Semester

21CSC201J – DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

PART – A (20 × 1 = 20Marks) Marks BL CO PO

Answer **ALL** Questions

1. How do you instantiate an array in C? 1 1 1 1
   (A) int Arr[ ] = New int (3); (B) int Arr [3];
   (C) int Arr[ ] = new int[3]; (D) int Arr( ) = New int [3];

2. Elements in an array are accessed 1 1 1 1
   (A) Randomly (B) Sequentially
   (C) Exponentially (D) Logarithmically

3. Which of the following data structure is non linear type? 1 1 1 3
   (A) Strings (B) Lists
   (C) Stacks (D) Hierarchical

4. What will be the output of the following code? 1 2 1 1

```
struct student | struct student *S ;
 {             | S→C = "HELLO";
  CHAR *C;     | printf("%S", S→C);
 };            |  }
 void main( )
 {
```

   (A) Hello (B) Segmentation fault
   (C) Runtime error (D) Nothing

5. Linked list data structure offers considerable saving in _____ 1 1 1 1
   (A) Computational time (B) Space utilization
   (C) Space utilization and computational time (D) Speed utilization

6. Which of the following is false about a doubly linked list? 1 1 2 1
   (A) We can navigate in both the directions (B) It requires more space than a singly linked list
   (C) The insertion and deletion of a node take a bit longer time (D) Implementing a doubly linked list is easier than singly linked list

7. Which of the following is false about a circular linked list? 1 1 2 1

(A) Every node has a successor
(B) Time complexity of inserting a new node at the head of the list is O(1)
(C) The complexity for deleting the last node is O(n).
(D) We can traverse the whole circular linked list, starting from any point

8. Which of these is not an application of a linked list? 1 2 2 2

(A) To implement file system
(B) For separate chaining in hash-tables
(C) To implement non-binary trees
(D) Random access of elements

9. The data structure required to check whether an expression contains balanced parenthesis is 1 1 3 3

(A) Stack
(B) Queue
(C) Array
(D) Tree

10. The prefix form of A-B / (C*D∧E) is? 1 2 3 3

(A) –/*∧ACBDE
(B) –ABCD * ∧DE
(C) –A/B*C∧DE
(D) –A/BC*∧DE

11. A queue follows _____ 1 1 3 3

(A) FIFO (First in First Out) principle
(B) LIFO (Last in First Out) principle
(C) Ordered array
(D) Linear tree

12. A normal queue, if implemented using an array of size max-size gets full when? 1 1 3 2

(A) Rear = Max – size – 1
(B) Front = (rear + 1) mod max size
(C) Front = Rear + 1
(D) Rear = Front

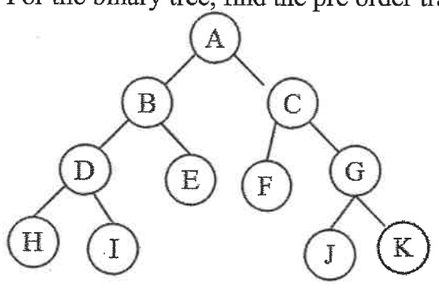
13. A linear list of elements in which deletion can be done from one end (front) and insertion can take place only at the other end (rear) is known as _____. 1 1 3 1

(A) Queue
(B) Stack
(C) Tree
(D) Linked list

14. What is a full binary tree? 1 1 4 1

(A) Each node has exactly zero or two children
(B) Each node has exactly two children
(C) All the leaves are at the same level
(D) Each node has exactly one or two children

15. For the binary tree, find the pre order traversal sequence. 1 1 4 3



(A) H I D E G F J K G C A
(B) H D I B E A F C J G K
(C) A B D H I E C F G J K
(D) A B C D E F G H I J K

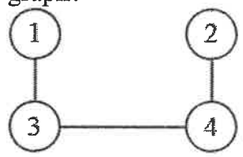
16. What is direct addressing? 1 1 4 1

(A) Distinct array position for every possible
(B) Fewer array positions than keys
(C) Fewer keys than array positions
(D) Same array positions for all keys

17. What data organization method is used in Hash tables? 1 1 4 3

(A) Stack
(B) Array
(C) Linked list
(D) Queue

18. What would be the number of zeros in the adjacency matrix of the given graph? 1 1 5 3



(A) 10
(B) 6
(C) 16
(D) 0

19. The traveling salesman problem can be solved using _____ 1 1 5 1

(A) A spanning tree
(B) A minimum spanning tree
(C) Bellman – Ford algorithm
(D) DFS traversal

20. DIJKSTRA's algorithm is used to solve ____ problem 1 1 5 1

(A) All pair shortest path
(B) Single source shortest path
(C) Network flow
(D) Sorting

PART – B (5 × 8 = 40 Marks)

Answer **ALL** Questions

	Marks	BL	CO	PO
21. a. Define a function to multiply two m × n matrices and return the resultant matrix.	8	4	1	9
(OR)				
b. How mathematical notations are used to measure the complexity of the algorithm? Explain different notations used.	8	3	1	3
22. a. Write an algorithm to demonstrate a polynomial equation using linked list for subtraction of two polynomials equations.	8	3	2	2
(OR)				
b. Consider an array A[1:n] Given a position, write an algorithm to insert an element in the array. If the position is empty, the element is inserted easily. If the position is already occupied the element should be inserted with the minimum number of shifts (Note: The elements can shift to the left or to the right to make the minimum number of moves).	8	4	2	3

Reg. No.															

B.Tech./M.Tech(Integrated) DEGREE EXAMINATION, JULY 2023

Third Semester

21CSC201J – DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year 2021-2022 & 2022-2023)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part – B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Answer **ALL** Questions

	Marks	BL	CO	PO
1. _____ is a variable that stores the address of another variable. (A) Local variable (B) global variable (C) auto variable (D) pointer variable	1	1	1	1
2. What is the best case time complexity of Linear search? (A) O(n) (B) O(1) (C) O(nlogn) (D) O(logn)	1	1	1	1
3. Which of the following data structure is non linear type? (A) string (B) list (C) Tree (D) Stack	1	1	1	3
4. Elements in an array are accessed _____ (A) randomly (B) sequentially (C) exponentially (D) logarithmically	1	1	1	1
5. The primary benefit of the Linked List data structure is _____ (A) computation time (B) space utilisation (C) space utilisation and computation time (D) speed than array ds	1	1	2	1
6. Find the correct code for the following node structure. Prev* | Data | Next* (A) struct node {struct node *prev;int data;struct node *next;} (B) struct node { struct node prev; int data; struct node next;} (C) struct node { int *prev; int data; int *next;} (D) struct node { struct node *prev; struct node data; struct node *next;}	1	2	2	1
7. Find the concept which is right for the below condition. "Any node can be a starting point. We can traverse the whole list by starting from any point. We just need to stop when the first visited node is visited again". (A) Singly linked list (B) array (C) Circular linked list (D) stack	1	2	2	1

8. _____ is a collection of similar type of data items. 1 1 2 1

(A) array (B) table
(C) record (D) structure

9. Which data structure follows LIFO concept? 1 1 3 3

(A) queue (B) array
(C) stack (D) deque

10. The result evaluating the postfix expression 5 5+60 6/*10- is _____ . 1 2 3 3

(A) 110 (B) 90
(C) 80 (D) 60

11. In Queues, we can insert an element at ___ end and can delete an element at _____ end. 1 1 3 3

(A) REAR,FRONT (B) FRONT,REAR
(C) TOP,BOTTOM (D) BOTTOM, TOP

12. How many queues are needed to implement a stack Data Structure? Consider there is no other Data Structure available like array and linked list. 1 2 3 3

(A) 1 (B) 3
(C) 2 (D) 4

13. While Inserting the elements 71, 65, 84, 69 in an empty Binary Search Tree (BST) in the sequence shown, the element in the lowest level is _____. 1 2 4 3

(A) 65 (B) 84
(C) 69 (D) 71

14. An AVL tree is a self – balancing binary search tree, in which the heights of the two child sub trees of any node differ by _____ 1 2 4 3

(A) Atleast one (B) 0
(C) Exactly one (D) Atmost one

15. What is full binary tree? 1 1 4 3

(A) Each node has exactly zero or two children (B) Each node has exactly two children
(C) All the leaves are at the same level (D) Each node has exactly one or two children

16. If "in order" traversal is performed in BST, then the resultant is _____. 1 2 4 3

(A) Elements are in non-increasing order (B) Elements are in decreasing order
(C) Elements are in increasing order (D) Elements are in random order

17. A graph with all vertices having equal degree is known as a _____. 1 1 5 3

(A) Multi Graph (B) Complete Graph
(C) Simple Graph (D) Regular Graph

18. Which data structure is used to traverse the graph by using Breadth First Search 1 2 5 3

(A) stack (B) queue
(C) Array (D) List

19. _____ is a graph with no cycles. 1 1 5 3

(A) weighted graph (B) cyclic graph
(C) acyclic graph (D) connected graph

20. Identify the data structure that performs insertion, deletion with an average time complexity of O(1). 1 2 5 4

(A) stack (B) hash table
(C) List (D) queue

PART – B (5 × 8 = 40 Marks)
Answer **ALL** Questions

	Marks	BL	CO	PO
21. a. Write a program to add two n×n matrices and return the resultant matrix.	8	4	1	9
(OR)				
b. Write pseudo code to perform result processing of a student. The student structure contains the following fields. i) name ii) no iii) Mark1, Mark2, Mark3	8	4	1	9
22. a.i. Write the advantages and disadvantages of Linked List over array.	4	3	2	3
ii. Write a pseudocode to insert an element in an array at the end.	4	3	2	3
(OR)				
b. Discuss and demonstrate an algorithm for the following operations in a singly linked list. i) Insert an element at the end ii) Delete an element from beginning	8	3	2	9
23. a. Write an algorithm for Push and Pop operations on Stack using array implementation.	8	3	3	3
(OR)				
b. Explain the addition and deletion operations performed on a circular queue with necessary algorithms.	8	3	3	3
24. a. Write the purpose of Rotation operation in AVL tree. Explain the types of rotations with example.	8	1	4	3
(OR)				
b. Write an algorithm for inserting and deleting a node in a binary search tree with example.	8	1	4	3
25. a. Explain the various representation of graph with your own example in detail.	8	5	5	2
(OR)				
b. Describe Minimum Spanning Tree Problem. Explain how to determine minimum cost spanning tree with an example (Use Kruskal's Algorithm).	8	5	5	2

PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Questions

	Marks	BL	CO	PO
26. Simulate the call logs in the mobile using an appropriate data structure. The Call Logs can store the telephone number, name of the 10 most recent callers. Once the limit of 10 is reached, and another call is made, the least recent number is deleted to make room for the recent number. (Illustrate with your own example)	15	5	3	3
27. Compute the contents of a hash table of 5 entries using separate chaining method, how to insert the following data 28, 18, 13, 20, 25.Show all computation clearly. Also use Linear probing method to handle collision technique.	15	5	4	3

* * * * *